

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH, MOHALI
END-SEMESTER 2025-2026

CHM201: SPECTROSCOPIC AND OTHER PHYSICAL METHODS

FULL MARKS 50

DURATION 3 HR

Constants: $N = 6.023 \times 10^{23} \text{ mol}^{-1}$; $k = 1.381 \times 10^{-23} \text{ JK}^{-1}$; $h = 6.626 \times 10^{-34} \text{ Js}$; $c = 2.998 \times 10^8 \text{ ms}^{-1}$; $R = 8.314 \text{ J.K}^{-1}/\text{mol.}$; $1 \text{ cm}^{-1} = 11.958 \text{ J mol}^{-1}$; $1 \text{ eV} = 8065.5 \text{ cm}^{-1}$

- 1) A molecule absorbs photons at 16310 cm^{-1} , and subsequently emits at 14190 cm^{-1} . i) what kind of spectroscopy records this? ii) Describe with diagram how the energy of the incident photon is conserved. iii) Where does that energy go? $1 + 2 + 2 = 5$
- 2) A molecule absorbs a photon at $\lambda = 180 \text{ nm}$. The molecule is ionized and an electron is emitted with velocity of $4.1 \times 10^5 \text{ m.s}^{-1}$. what kind of spectroscopy is this? what is the binding energy of the emitted electron? $1 + 2 = 3$
- 3) **Schematically** demonstrate the number of electronic transitions possible from ns^2 to $ns^1(n+1)p^1$ states in absence and presence of weak magnetic field. Please cite the relative energy differences clearly for a weak magnetic field of B_0 . Marks will be based on scheme or diagram and write the energy differences (ΔE) on the scheme or diagram (Hint. $g = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}$). $3 + 4 = 7$
- 4). Several possible transitions for beryllium atom are listed below. Are the term symbols in parenthesis valid? A. Write why valid. B. Which are the transitions "fully allowed"? Cite reason from selection rule in one sentence for each point. Everything should in one table. $4 \times 2 = 8$

Transitions	Term Symbol Valid?	Reason	Transitions Allowed	Reason
$2s5s (^1S_0) \rightarrow 2s5d (^1D_2)$				
$2s5s (^3S_1) \rightarrow 2s2p (^1P_1)$				
$2s5p (^3P_1) \rightarrow 2s3s (^3S_1)$				
$2s3p (^3P_1) \rightarrow 3p2 (^3D_2)$				

- 5) Arrange the Terms of d^8 according decreasing energy. 2
 $^3F_3; ^3P_1; ^3F_2; ^3P_0; ^1G_4; ^1D_2; ^1S_0; ^3F_4; ^3P_2$

- 6) The table below lists the IR/Raman activities for cyanogen (C_2N_2). Assign the vibrational modes for each spectral line. a) Fill column 4. $5 + 1 + 2 = 8$

Frequency (cm ⁻¹)	Raman	IR	Assignments of vibrational modes
2322	Strong		
2149		strong, PR contour	
848	strong		
506	Very strong		
226		strong, PQR contour	

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$2s^1 3p^1$
 $S = 1, 0$
 $L = 1 + 0 = 1$
 $J = 2, 1, 0$
 $^3P_2, ^3P_1, ^3P_0$

$3p^2$
 $S = 2, 1, 0$
 $L = 2$
 $J = 2, 1, 0$
 $^3P_2, ^3P_1, ^3P_0$
 1D_2

$^3F_4 < ^3F_3 < ^3F_2 < ^3P_2 < ^3P_1 < ^3P_0 < ^1G_4 < ^1D_2 < ^1S_0$

b. Draw the Structure here of C_2N_2 .

c. The wavenumber of the incident radiation in a Raman spectrometer is 20487 cm^{-1} . What is the wavenumber of the scattered Stokes radiation for the $J = 2 \leftarrow 0$ in $^{15}N_2$ ($B = 1.9896\text{ cm}^{-1}$)?

7. The rotational terms of a diatomic molecule (the energy levels expressed as wavenumbers) are given to a good approximation by:

$$\epsilon_J = BJ(J+1) - DJ^2(J+1)^2$$

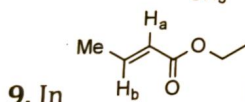
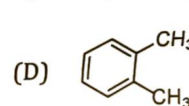
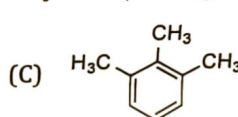
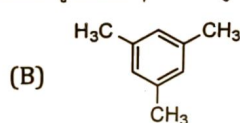
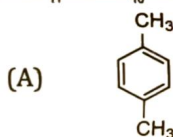
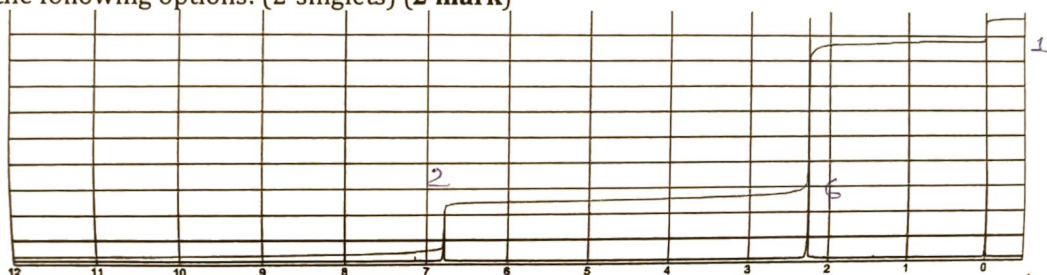
a) Write the meaning/name of the terms and their expressions for B and D.

b) By applying selection rules for pure rotational spectroscopy, derive an expression for the energy of transitions observed in a high-resolution rotational spectrum.

c) In a high-resolution microwave study of $^2H^{19}F$, the consecutive four lines in the spectrum were observed at: (i) 22.0180 cm^{-1} ; (ii) 44.0218 cm^{-1} ; (iii) 65.9970 cm^{-1} ; (iv) 87.9295 cm^{-1} . Deduce the values of J for each transition, B and D for $^2H^{19}F$.

d) Can we deduce J using the above transitions, if we only use energy expression for rigid-linear rotor? Please justify your answer in one sentence.

8. Based on the following 1H -NMR spectrum in $CDCl_3$, identify the correct compound among the following options: (2-singlets) (2 mark)



(A) $\delta_{Hb} > \delta_{Ha}$ due to mesomeric effect

(C) $\delta_{Hb} > \delta_{Ha}$ due to inductive effect

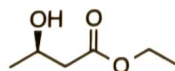
(B) $\delta_{Ha} > \delta_{Hb}$ due to inductive effect

(D) $\delta_{Ha} > \delta_{Hb}$ due to mesomeric effect

10. The 1H -NMR of CH_3F showed a doublet due to coupling of ^{19}F . A 100 MHz NMR instrument has been used for recording the spectra. If the doublet signals appear at 4.525 and 4.075 ppm, the coupling constant (J) will be (1 mark)

(A) 45 Hz (B) 4.5 Hz (C) 0.0045 Hz (D) 0.45 Hz

11. Predict the 1H -NMR spectral features (number of signals and splitting pattern) of the following molecule: (Note: Consider that OH proton is not involved in the coupling pattern) (3 marks)



$$\begin{aligned} & B(J+2)(J+3) - BJ(J+1) \\ & = B[J^2 + 5J + 6 - J^2 - J] \\ & = B(4J + 6) \end{aligned}$$